

1 Data

iSAID: human masks
SAMRS: SAM-derived
Plane + Small Vehicle
512 images each



2 Localization

GT bbox builds references
YOLO bbox tests models
Same image and instance
for every reference



3 Frozen models

SAM1 / SAM2 / SAM3
No fine-tuning
Predictions reused for
every reference mask



4 Scoring

Fixed prediction versus
baseline + SAM1/2/3
Instance-macro IoU
No test-time changes

**Controlled comparison: image, instance, bbox and prediction stay fixed.
Only the evaluation mask changes.**

Primary quantity

Extra IoU on own label = IoU on the model's own label - mean IoU on labels from the other two SAM models

Interpretation Positive value: the fixed prediction scores higher with labels made by its own model.